Victor Barreto Andrade Gonçalves

Curriculum Vitae · Applicant for the skill **Development and Implementation of Industrial AI Solutions** — BRICS Hi-Tech Future Skills Competition 2026

Year of birth	2006	**Citizenship**	Brazil
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PROFILE

Computer Engineering student at SENAI CIMATEC University (Brazil). I build AI that works on real systems: computer vision on an autonomous inspection robot, deep learning on a GPU cluster, AI agents deployed with safety checks and fallback logic, and a simulated cobot sorting cell that rebuilds the four competition modules of this skill.

EDUCATION

University	SENAI CIMATEC University (Universidade SENAI CIMATEC), Salvador, Brazil
Faculty / program	Engineering — Computer Engineering program (5-year degree)
Specialty and degree	Computer Engineering, Bachelor of Engineering (B.Eng.)
Years of study	January 2024 – December 2028 (expected graduation); currently 3rd year (6th of 10 semesters)

WORK EXPERIENCE

High Performance Computing (HPC) Intern — Oct 2025 – present
SENAI CIMATEC, Salvador, Brazil

- Apply deep learning models (PyTorch, TensorFlow) to speed up molecular simulations and predictive models for drug discovery.
- Run and optimize model training on a GPU cluster managed by SLURM (parallel jobs, GPU allocation); process large datasets with Python, pandas and SQL.

IEEE Robotics and Automation Society (RAS), Student Chapter — Mar 2024 – present (volunteer)
SENAI CIMATEC

- Treasurer (2025 – present): budget planning and fundraising through IEEE grants and sponsors. Earlier roles: Trainee on Leak Track (2024) and Project Advisor of the X-ray project (2025 – 2026), both described below.

Campus student leader selected by Google and by AWS — Nov 2025 – present (volunteer)
SENAI CIMATEC

- Google Student Ambassador (Aug 2026 – present): designed and ran an AI competition on Google Skills during SEMCOMP 2026, where participants built prototypes in Google AI Studio and earned an official badge.
- AWS Student Builder Group Leader, formerly Cloud Club Captain (Nov 2025 – Jul 2026): led the campus cloud community with workshops on AWS and AI.

Competitive Programming Club — Nov 2024 – present (volunteer)
SENAI CIMATEC

- Teaching monitor (2024 – 2025): taught C++ algorithms; the club's 4 teams placed 2nd, 5th, 8th and 11th of 22 at the regional ICPC-format contest. Problem setter for the 2nd Cimatec Computing Olympiad (OCC), October 2026.

ACHIEVEMENTS RELATED TO THIS SKILL

CobotSort AI — simulated collaborative-robot sorting cell (personal project) — October 2026

- Rebuilt the four competition modules in simulation (MuJoCo, UR5e 6-axis cobot as a stand-in for the ARM95): PyQt6 GUI for manual and semi-automatic sorting, fully automatic sorting, YOLO11n object detection trained on synthetic data, and its integration into the cell. Code: github.com/VictorBarretoAndrade/BRICS_CobotSort
- Kinematics and safety: DH forward kinematics, multi-seed inverse kinematics (100/100 targets solved, error < 0.01 mm), singularity blocking by manipulability index, collision checks, E-stop and signal lamp; **10/10 unsafe requests blocked**.
- Results over 20 automatic episodes (176 cubes), reported in the competition's order: **0 unplanned contacts**, 97.2% of cubes sorted without the operator (the rest stayed because their bin was full), 0% wrong bin, **0.91 mm mean placement error** with YOLO. Detector mAP@50: 0.991 on an out-of-distribution test set vs. 0.942 for a classical HSV baseline.
- Operator agent with the EduBot pattern: orders in Portuguese or English become validated tool calls; 10/10 evaluation orders passed, including refusing a non-existent bin and asking back on ambiguous orders.

Leak Track — autonomous industrial inspection robot IEEE RAS · Mar – Dec 2024

- Co-developed a robot that detects leaks in industrial structures that are hard to reach or dangerous for human inspectors: ROS 2 control, Python and OpenCV image processing.
- Real-time CNN analysis of images from a FLIR thermal camera to find temperature changes linked to leaks.

EduBot — AI agent deployed in a full software system Project Lead and Tech Lead · 2026

- Led the 8-person team as project lead and tech lead: defined the architecture and integrated back end, front end and AI layer. **EduBot took the team to the HACKaSTONE 2026 Grand Final**, a global contest on AI for education (Amsterdam), and is the subject of my lead-author paper accepted at the Conference AI for Education The Netherlands 2026 (see Publications).
- AI agent on the Anthropic API (Claude via AWS Bedrock) that diagnoses a student's weakest skill with four controlled tools; input validation, guardrails against invented data, and a rule-based fallback if the model fails.
- Python/Flask REST API (67 endpoints), MySQL (31 tables), React + TypeScript, Docker Compose, 332 automated tests.

Computer vision for chest X-ray diagnosis IEEE RAS · Jan 2025 – Jan 2026

- As Project Advisor, guided a multidisciplinary team building a CNN system that separates healthy lungs from pneumonia and tuberculosis cases, from data curation and validation to a results interface. Next stage with new trainees: object detection (Faster R-CNN) on the RSNA dataset.

AWS DeepRacer — reinforcement learning for an autonomous vehicle 2025

- Trained reinforcement learning models to drive an autonomous race car: **6th place in the Brazilian National Final** (Brazilian Computer Society Congress, CSBC 2025); 3rd place in the Northeast semifinal.

PUBLICATIONS

- **Lead author.** “What Instantiation Reveals: Building an Agentic Remediation Loop on Top of a Conceptual Model for AI-Enhanced Virtual Learning Environments.” Accepted for presentation at the Conference AI for Education The Netherlands 2026, Amsterdam. The paper presents EduBot as the instantiation of the model.
- **Co-author.** Vilas Boas JB, Rodrigues RSC, Gonçalves VBA, Silva YQJ, Johnsson MLN, Lima CSPC. “CatchNet: Specification and Prototyping of a Decision-Support Application for Non-Technical Loss Management in Power Distribution Networks.” Poster, SIINTEC 2026 (International Symposium on Innovation and Technology, SENAI CIMATEC). Application for electricity utilities, designed to integrate an ML model that scores theft probability.

OTHER AWARDS AND SELECTIONS

- **2nd place of 22 teams, Salvador site** — Brazilian Programming Marathon of the Brazilian Computer Society (SBC, ICPC sub-regional), 2026; Honorable Mention in 2024 and 2025.
- **1st place** — SEMCOMP 26 Programming Marathon, Federal University of Bahia (UFBA), 2026.
- **3rd place in Bahia** — 1st Northeastern Brazil Programming Contest (ICPC format), 2026.
- **2nd place** — Scientific Article Competition, Researcher Challenge Seminar (Seminário Desafio Pesquisador), SENAI CIMATEC.
- **1 of 28 students selected in Brazil** — 9th HackLab FNESP 2026, innovation marathon (Semesp).

TECHNICAL SKILLS BY AREA OF THE SKILL

Area	Skills and tools	Where I used them
Collaborative industrial robotics	ARM95 cobot programming (waypoints, joint and linear moves, vacuum tool); forward and inverse kinematics; DH parameters; singularities; cobot safety; MuJoCo simulation	CobotSort; Skills Camp 2026
Programming and control	Python, C++, SQL, TypeScript; ROS 2; Git, GitHub, GitLab; REST APIs (Flask); user interfaces (PyQt6, React)	CobotSort; Leak Track; EduBot
Neural networks and computer vision	PyTorch, TensorFlow, Keras; CNNs; OpenCV; thermal imaging (FLIR); object detection (YOLO, Faster R-CNN); data augmentation; precision, recall, mAP	CobotSort; Leak Track; X-ray project; HPC internship
AI implementation and deployment	Docker, Linux, GPU clusters (SLURM), AWS; LLM agents with tool use; guardrails and fallback logic; automated testing	CobotSort; EduBot; HPC internship

LANGUAGES AND CERTIFICATIONS

- **Languages:** Portuguese (native), English (C2 Proficient, EF SET English Certificate).
- **Certifications:** Microsoft Certified: Azure AI Fundamentals.